

Serverless Computing

Reducing Infrastructure Management in
the Cloud

Technical Presentation

CSA1506-Cloud Computing and virtualization

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What is Serverless Computing?

A cloud computing model where the cloud provider dynamically manages the allocation and provisioning of servers, allowing developers to focus purely on writing code.

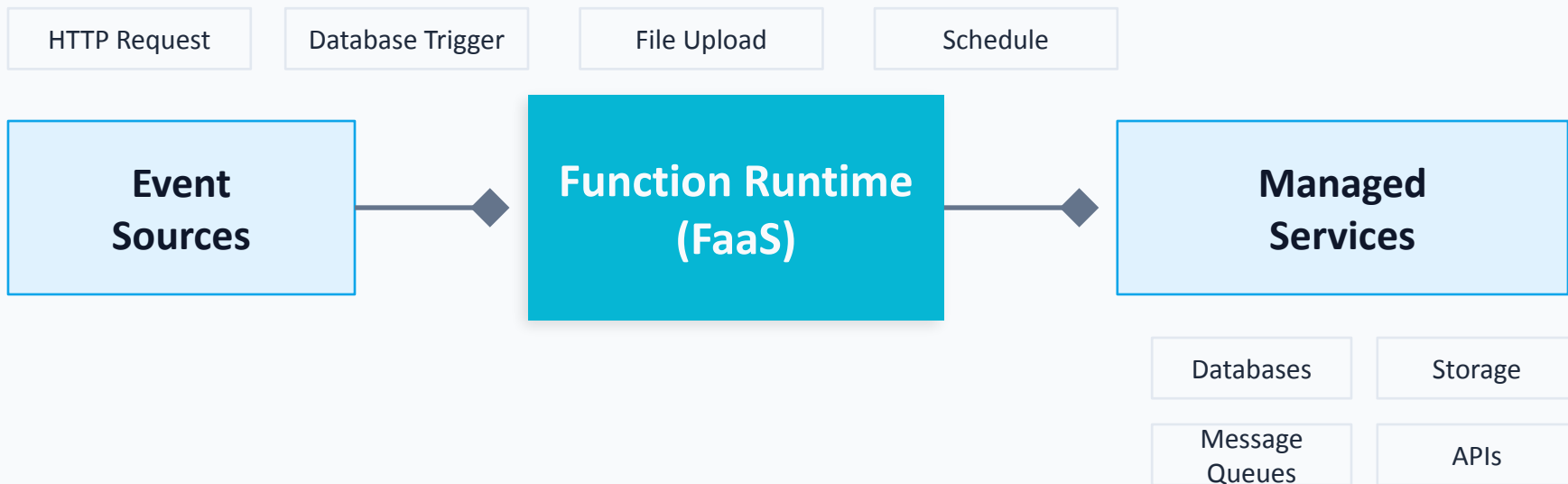
Function-as-a-Service (FaaS)

- Event-driven compute service
- Run code in response to triggers
- Pay only for execution time

Leading FaaS Platforms

- AWS Lambda
- Azure Functions
- Google Cloud Functions
- Cloudflare Workers

Serverless Architecture



The provider handles provisioning, scaling, patching, and availability automatically.

Business Advantages

Reduced Operational Cost

No idle server costs; pay only for actual compute time consumed.

Automatic Scaling

Scale from zero to thousands of requests instantly without manual configuration.

Faster Time-to-Market

Developers deploy features in hours instead of weeks.

Focus on Business Logic

Eliminate infrastructure management, patching, and capacity planning.

Practical Use Cases

1

Web & Mobile Backends

APIs and microservices that scale with user demand.

2

Data Processing

ETL pipelines, stream processing, and real-time analytics.

3

Automation

Scheduled tasks, notifications, and DevOps workflows.

4

IoT & Event Handling

Process device telemetry and respond to events in real time.

Key Takeaways

- Serverless abstracts away server management, letting teams focus on code.
- FaaS is the core compute model, driven by events and auto-scaling.
- It reduces costs through fine-grained, usage-based billing.
- Ideal for event-driven, variable, and microservice workloads.



Thank you